

TalentFlow - AI-Powered Hiring & Candidate Assessment Platform

SUMMARY

TalentFlow is a modern Applicant Tracking System (ATS) and Candidate Assessment Platform designed for small and medium-sized businesses that need a structured hiring process without the complexity and cost of enterprise recruitment software.

The platform centralizes candidate management, job postings, interview scheduling, assessments, analytics, and AI-powered recruitment assistance.

PROJECTS

TalentFlow - AI-Powered Hiring & Candidate Assessment Platform | A modern Applicant Tracking System (ATS) and Candidate Assessment Platform for SMBs to centralize recruitment workflows.

- Centralizes candidate management, job postings, interview scheduling, assessments, analytics, and AI-powered recruitment assistance.
- Core modules include authentication & authorization, job management, candidate management, application tracking, recruitment pipeline, technical assessments, interview management, notifications, and analytics dashboard.
- Implements admin capabilities such as user management, permission management, audit logs, and system settings.
- Designed with Clean Architecture and CQRS, using patterns including Repository, Unit of Work, Specification, Domain Events, and Factory.
- Recommended stack includes Angular frontend, ASP.NET Core Web API backend, SQL Server database, Hangfire background jobs, and Azure Blob Storage for production.
- Defines API endpoints for auth, jobs, candidates, applications, interviews, and assessments.
- Includes AI features: resume parsing, job matching, candidate ranking, interview question generation, feedback summarization, and job description generation.
- Planned roadmap spans MVP, Professional, AI, and SaaS phases with multi-tenant architecture and subscription capabilities.

TECHNICAL SKILLS

Angular

Angular Material

RxJS

NgRx (Optional)

SignalR Client

ASP.NET Core Web API

Entity Framework Core

FluentValidation

AutoMapper

MediatR

Serilog

SQL Server

Hangfire

Local File Storage

Azure Blob Storage

Clean Architecture
CQRS
Repository Pattern
Unit of Work
Specification Pattern
Domain Events
Factory Pattern
JWT Authentication
Refresh Tokens
Role-Based Access Control

PROBLEM STATEMENT

Many companies still manage hiring through spreadsheets, emails, and disconnected tools.

This creates problems such as: - Lost candidate information - Poor hiring visibility - Inconsistent interview processes - Slow recruitment cycles - Difficult reporting and analytics

TalentFlow solves these issues by providing a complete recruitment workflow in a single platform.

TARGET USERS

Primary Users: - HR Specialists - Recruiters - Hiring Managers - Startup Founders

Secondary Users: - Candidates - Technical Interviewers

BUSINESS GOALS

- Reduce hiring time - Improve candidate evaluation - Centralize recruitment operations - Provide hiring insights and analytics - Automate repetitive recruitment tasks

DEVELOPMENT ROADMAP

Phase 1 - MVP (3-4 Weeks): Authentication, Roles, Jobs, Candidates, Applications, Pipeline Board, Interviews, Dashboard.

Phase 2 - Professional Version (2-3 Weeks): Assessments, Notifications, Reports, Audit Logs, File Uploads, Hangfire Jobs.

Phase 3 - AI Version (2-4 Weeks): Resume Parsing, Skill Extraction, Match Scoring, Candidate Ranking, Interview Question Generation, Feedback Summaries.

Phase 4 - SaaS Version (4-6 Weeks): Multi-Tenant Architecture, Subscription Plans, Company Workspaces, Tenant Isolation, Usage Analytics.

TECHNICAL CHALLENGES

- Designing a flexible recruitment workflow - Role-based authorization - Complex filtering and searching - CQRS implementation - Multi-tenant data isolation - Real-time notifications - AI integration - Scalable database design - Performance optimization

WHY THIS PROJECT IS STRONG FOR A PORTFOLIO

This project demonstrates: - Real business value - Enterprise architecture - Clean Architecture - CQRS - SQL Server design - Authentication & Authorization - Angular frontend development - Complex workflows - Reporting & analytics - AI integration - SaaS scalability

Unlike typical CRUD projects, TalentFlow showcases system design, business thinking, scalability, security, and engineering maturity that interviewers expect from a professional .NET developer.